

SPRINGFIELD COMANCHE, CO, USA

WMO: 724646

Lat: 37.283N

Lon: 102.614W

Elev: 1336

StdP: 86.27

Time Zone: -7.00 (NAM)

Period: 01-19

WBAN: 03028

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-12.6	-20.2	0.7	-4.1	-17.9	0.9	-2.3	14.7	4.2	12.8	3.2	3.5	270	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	36.6	17.7	35.2	17.6	33.8	17.5	20.4	29.3	19.9	29.2	19.3	28.7	6.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	15.4	22.0	17.5	14.8	21.4	17.0	14.3	21.3	64.8	28.9	62.8	29.1	60.8	28.7	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.2	12.2	11.0	DB	-21.1	39.1	2.8	1.1	-23.1	39.9	-24.7	40.5	-26.3	41.1	-28.3	41.9	(4)
			WB	-21.5	21.7	2.7	0.7	-23.5	22.2	-25.0	22.6	-26.6	23.0	-28.5	23.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	0.9	1.6	7.0	11.1	16.3	22.5	25.1	23.7	19.7	12.4	6.1	0.9
	DBStd	10.00	5.67	6.09	5.62	5.06	4.87	3.79	2.76	2.93	4.04	5.24	5.53	5.87
	HDD10.0	1166	284	240	124	47	10	0	0	0	1	36	140	284
	HDD18.3	2819	541	469	351	219	96	8	1	2	32	191	368	542
	CDD10.0	2014	2	4	32	80	204	377	467	424	292	110	22	1
	CDD18.3	627	0	0	0	3	32	134	209	167	73	8	0	0
	CDH23.3	8526	1	5	48	179	652	1840	2624	1926	994	238	19	0
	CDH26.7	4217	0	0	4	40	246	964	1469	988	436	67	1	0
Wind	WSAvg	5.2	4.6	5.2	5.6	6.3	5.8	5.7	5.0	4.7	5.2	5.0	5.0	4.8
Precipitation	PrecAvg	397	13	9	22	28	55	53	71	66	24	24	13	9
	PrecMax	550	45	32	87	93	118	179	210	212	73	89	61	45
	PrecMin	223	1	0	0	5	9	11	8	12	0	0	0	0
	PrecStd	71	11	9	20	21	30	35	45	46	22	27	14	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	23.7	27.2	30.4	33.8	38.4	38.0	37.5	35.3	31.2	25.8	20.5
		MCWB	7.7	8.5	10.6	11.5	14.3	16.4	18.4	17.6	16.4	13.8	10.2	7.6
	2%	DB	17.8	19.6	24.6	27.6	31.7	36.2	36.8	35.8	33.1	28.5	23.1	17.3
		MCWB	6.4	6.9	9.5	11.0	14.0	16.5	18.3	18.1	16.3	12.8	9.3	6.3
	5%	DB	14.2	16.3	21.9	25.5	29.5	34.2	35.5	33.9	31.4	26.2	20.0	14.1
		MCWB	4.9	5.7	8.7	10.6	13.5	16.6	18.2	18.0	16.4	12.7	8.0	4.9
	10%	DB	11.2	12.7	18.6	22.6	27.3	32.5	33.9	32.3	29.2	23.0	16.5	11.1
		MCWB	3.6	4.2	7.3	9.9	13.5	16.7	18.3	18.1	16.1	11.7	6.8	3.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	9.2	12.3	14.6	18.2	20.5	21.2	21.4	19.9	17.4	11.5	8.6
		MCDB	19.4	20.4	22.4	22.4	26.2	29.2	29.9	29.3	28.8	22.6	21.2	19.0
	2%	WB	6.7	7.5	10.6	13.2	16.8	19.5	20.5	20.5	18.9	15.5	10.2	6.7
		MCDB	16.5	18.4	21.4	21.6	25.4	29.3	29.7	28.9	27.1	23.2	20.2	16.2
	5%	WB	5.3	6.0	9.3	11.9	15.8	18.8	20.0	19.9	18.0	13.9	8.8	5.3
		MCDB	13.9	15.8	19.7	20.5	24.7	28.2	29.8	28.5	26.1	21.7	19.0	13.5
	10%	WB	3.7	4.5	8.1	10.8	14.8	18.1	19.4	19.3	17.2	12.6	7.4	3.8
		MCDB	10.8	12.5	17.9	20.8	23.5	27.6	29.5	27.8	25.4	21.7	16.6	10.7
Mean Daily Temperature Range	5% DB	MDBR	15.6	15.7	16.9	16.3	16.3	16.2	15.7	14.9	15.5	16.3	16.2	14.8
		MCDBR	20.3	21.6	21.1	21.2	19.5	18.8	18.0	17.7	18.5	20.2	20.8	19.7
		MCWBR	11.2	11.2	9.6	9.0	7.4	6.0	4.9	5.0	6.2	8.5	9.9	10.9
	5% WB	MCDBR	19.5	20.5	19.6	17.7	17.0	16.1	15.3	14.8	15.1	16.4	19.1	18.6
		MCWBR	10.9	10.9	9.5	8.3	7.3	6.1	4.8	5.1	5.6	8.2	9.5	10.7
Clear-Sky Solar Irradiance	taub	0.245	0.253	0.278	0.305	0.323	0.346	0.363	0.347	0.318	0.283	0.258	0.246	
	taud	2.593	2.588	2.515	2.447	2.457	2.421	2.427	2.440	2.503	2.579	2.573	2.585	
	Ebn at Noon	973	1001	995	976	956	930	911	921	936	949	945	948	
	Edh at Noon	71	82	97	110	112	116	114	110	98	82	72	67	
All-Sky Solar Radiation	RadAvg	2.86	3.69	5.04	6.19	7.14	7.65	7.50	6.63	5.61	4.23	3.15	2.50	
	RadStd	0.21	0.32	0.39	0.34	0.39	0.32	0.27	0.29	0.28	0.36	0.23	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+72	N/A

Nomenclature: See separate page